

635 Homework #10 Answers

$$1. F_z = 8\epsilon M_0 \cos(kd \cos\theta) \frac{e^{-jkr}}{4\pi r} \frac{1}{2} \frac{\sin(kl/2 \cos\theta)}{kl/2 \cos\theta}$$

$$\text{or} = (2m_0)\epsilon \cos(kd \cos\theta) \sin(kl/2 \cos\theta) \frac{e^{-jkr}}{\pi r k \cos\theta}$$

$$H_\theta = j\omega F_z \sin\theta$$

$$E_\phi = \eta H_\theta = j\omega \eta F_z \sin\theta = j \frac{k}{\epsilon} F_z \sin\theta$$

$$2. \psi_1^{TM} = \cos(k_x x) \sin(k_y y) e^{\pm jk_z z}, \quad k_y = \frac{m\pi}{b}, \quad m=1, 2, 3, \dots$$

$$\psi_2^{TM} = \cos[k_{x2}(a-x)] \sin(k_y y) e^{\pm k_z z}, \quad \text{no zero}$$

Transverse resonance model

$$\boxed{\frac{k_{x1}}{\epsilon_1} \tan(k_{x1} t) = \frac{k_{x2}}{\epsilon_2} \tan[k_{x2}(a-t)]} \quad (a)$$

$$\boxed{k_{x1}^2 - k_{x2}^2 = k_1^2 - k_2^2} \quad (b)$$

Neither k_y or k_z can be zero.

$$E_x = -\frac{j\omega}{k^2} (k^2 + \frac{\partial^2}{\partial x^2}) \psi$$

$$H_x = 0$$

$$E_y = -\frac{j\omega}{k^2} \frac{\partial^2}{\partial x \partial y} \psi$$

$$H_y = \frac{1}{\mu} \frac{\partial^2}{\partial z^2} \psi$$

$$E_z = -\frac{j\omega}{k^2} \frac{\partial^2}{\partial x \partial z} \psi$$

$$H_z = -\frac{1}{\mu} \frac{\partial}{\partial y} \psi$$

$$3. \psi_a = e^{-k_z z} e^{\pm jk_z z}, \quad x \geq t$$

$$\psi_b = \cos(k_x x) e^{\pm k_z z}, \quad 0 \leq x \leq t$$

$$(k_{xb} t)^2 + (\alpha_a t)^2 = (3.7)^2$$

$$\alpha_a t = k_{xb} t \cot(k_{xb} t)$$

$$k_{xb} t = 2.43, \quad \alpha_a t = 2.81$$